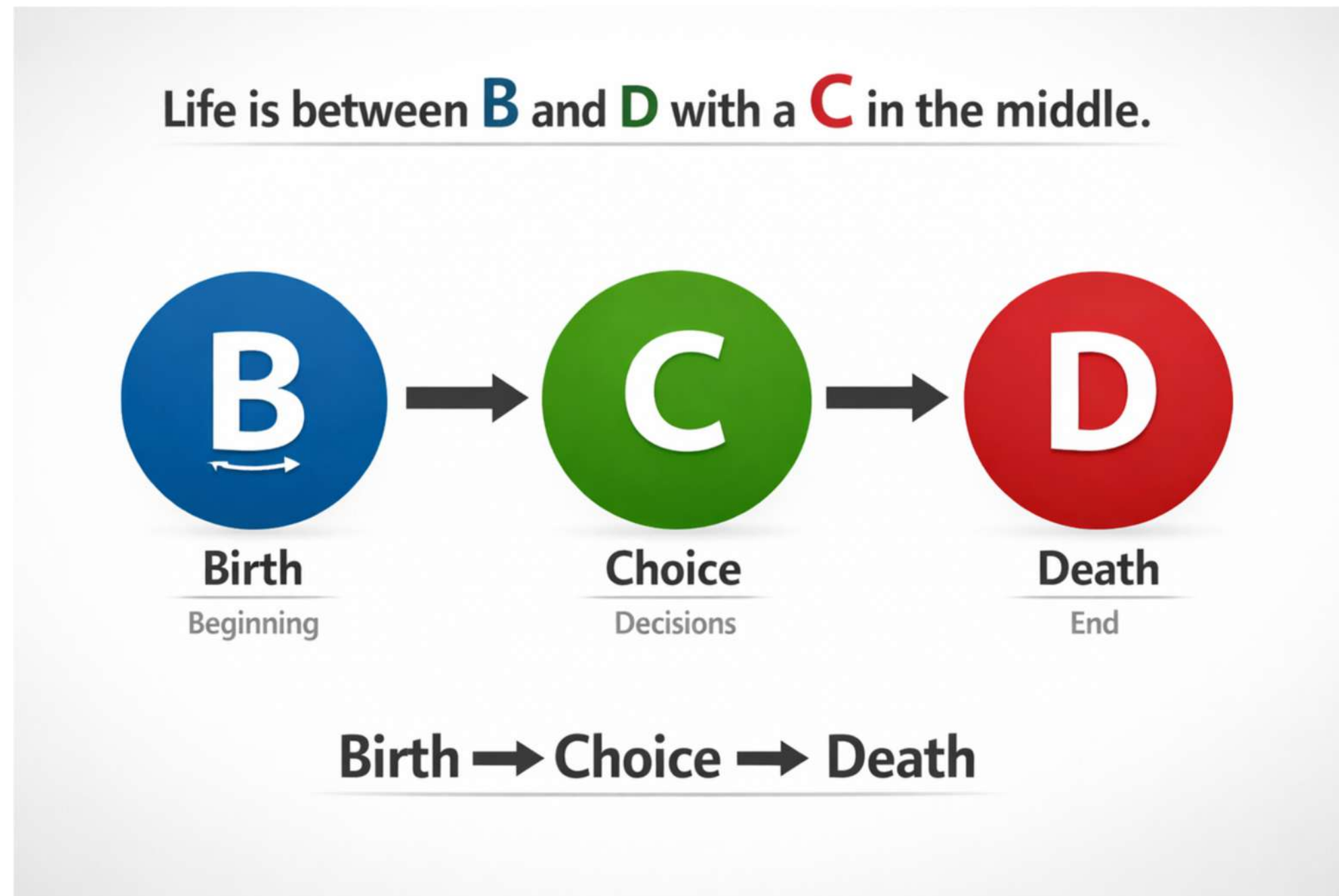

마시멜로와 벨만 방정식

R L 은 어떻게 미래 기대 가치를 현재로 가져오는가

김승환

마시멜로와 벨만 방정식

인생이란...



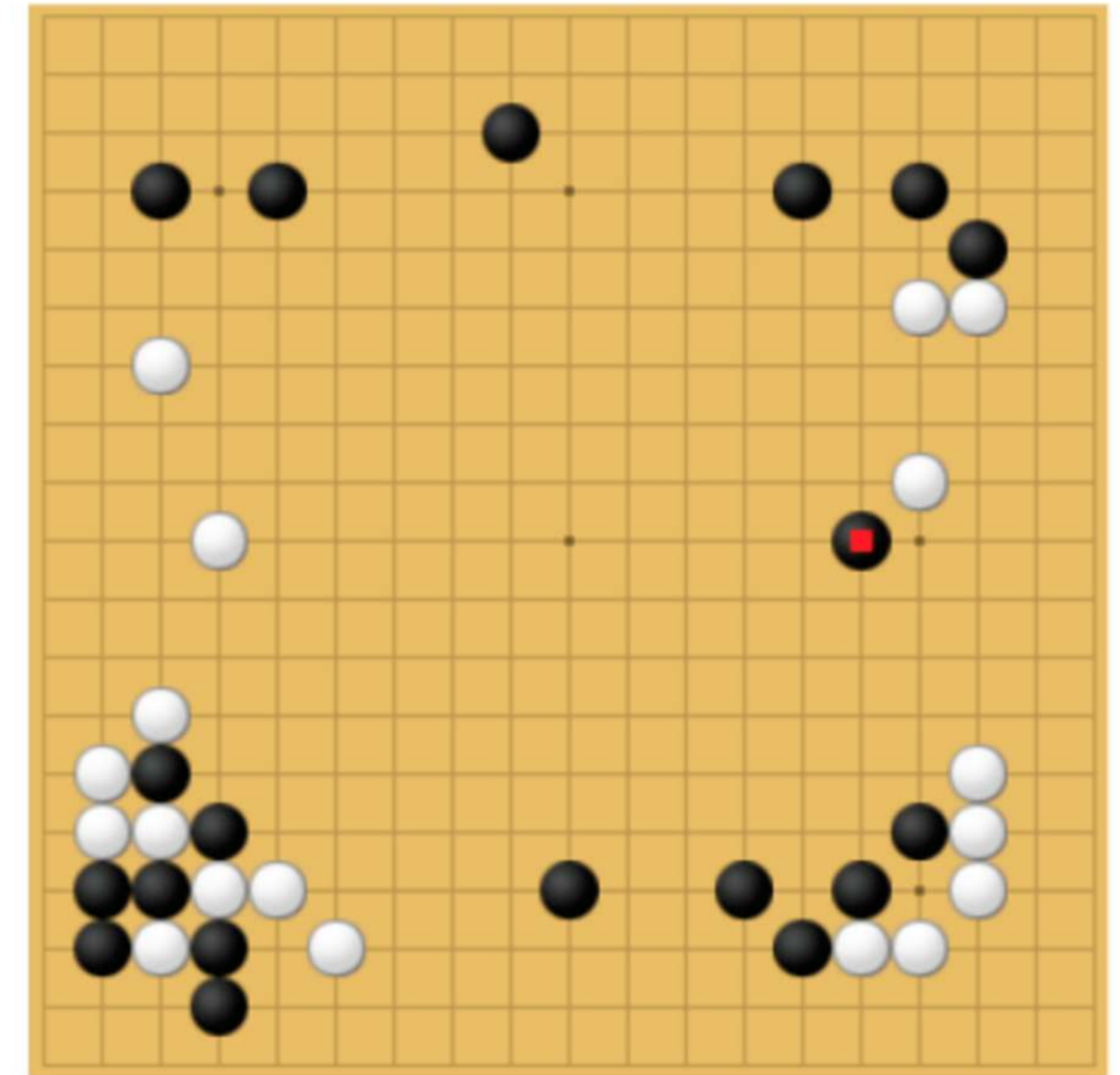
마시멜로와 벨만 방정식

마시멜로 실험



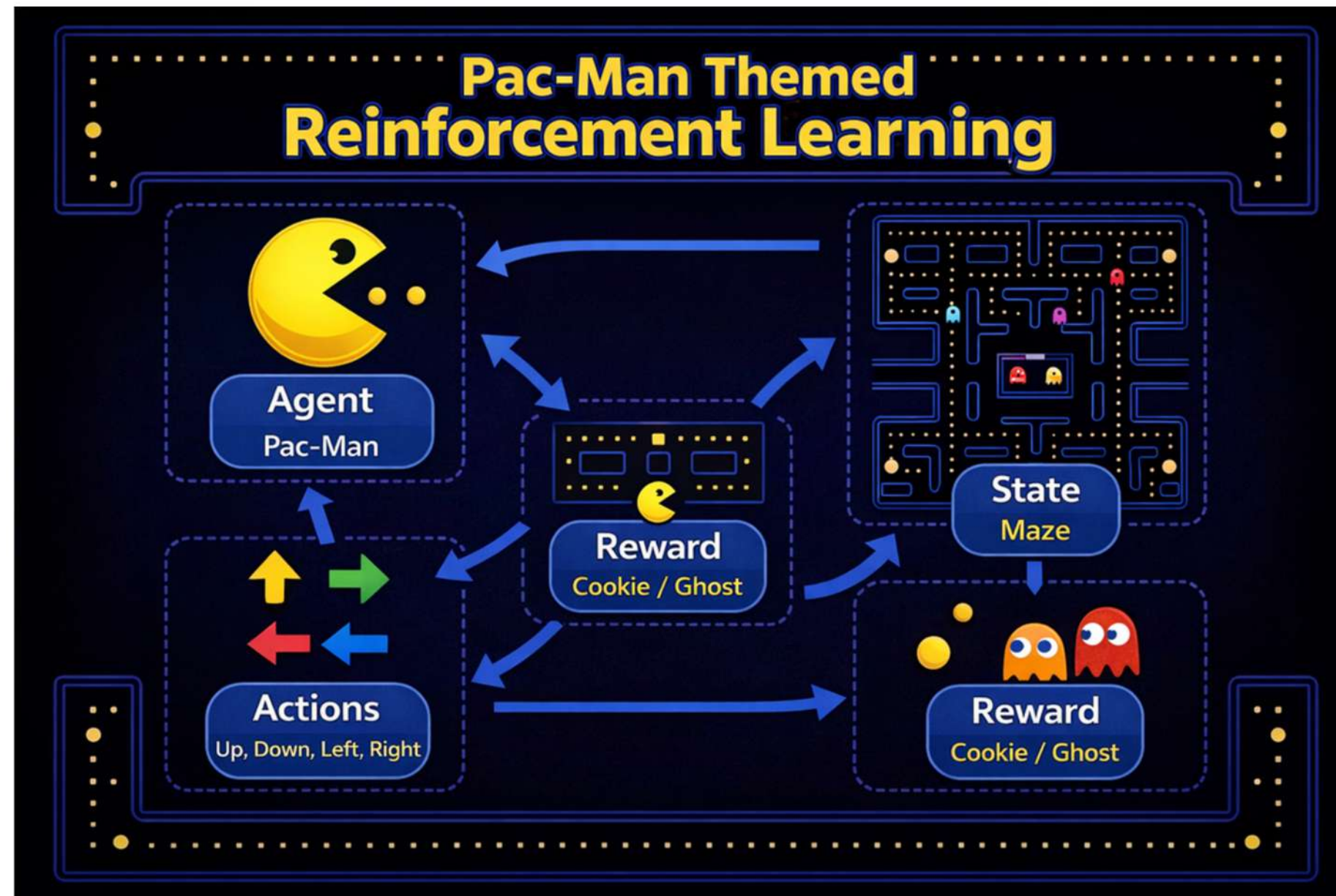
마시멜로와 벨만 방정식

알파고의 신의 한수



마시멜로와 벨만 방정식

1. 세상을 0과 1로 모델링하다 (Markov Decision Process)



마시멜로와 벨만 방정식

2. 가치를 측정하는 두 가지 눈

바둑판 전체를 평가하는 눈

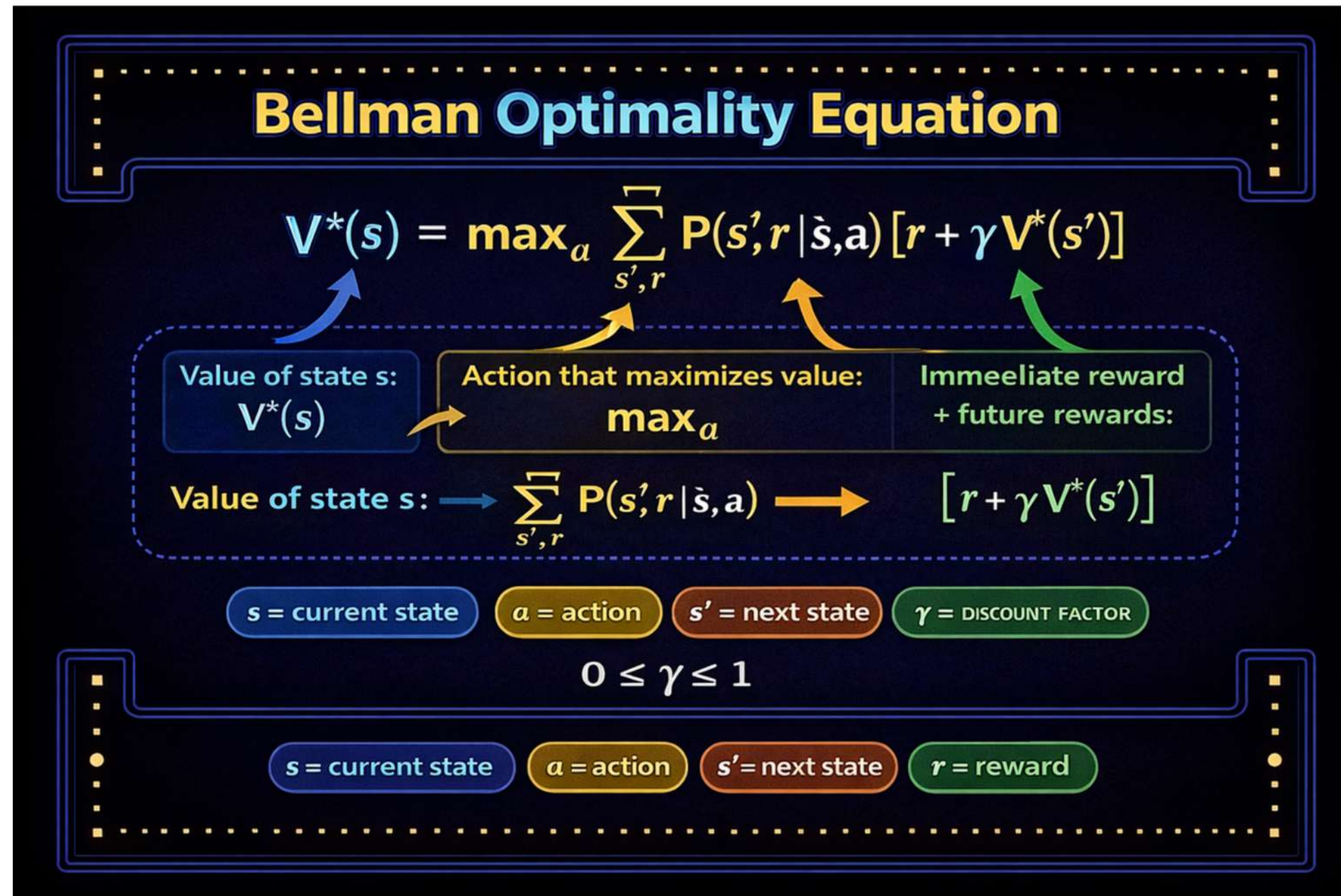
$$v_*(s) = \max_a \mathbb{E}[r + \gamma v_*(s') \mid s_t = s, a_t = a]$$

특정 위치에 돌을 놓았을 때의 결과를 평가하는 눈

$$q_*(s, a) = \mathbb{E}[r + \gamma \max_a q_*(s', a') \mid s_t = s, a_t = a]$$

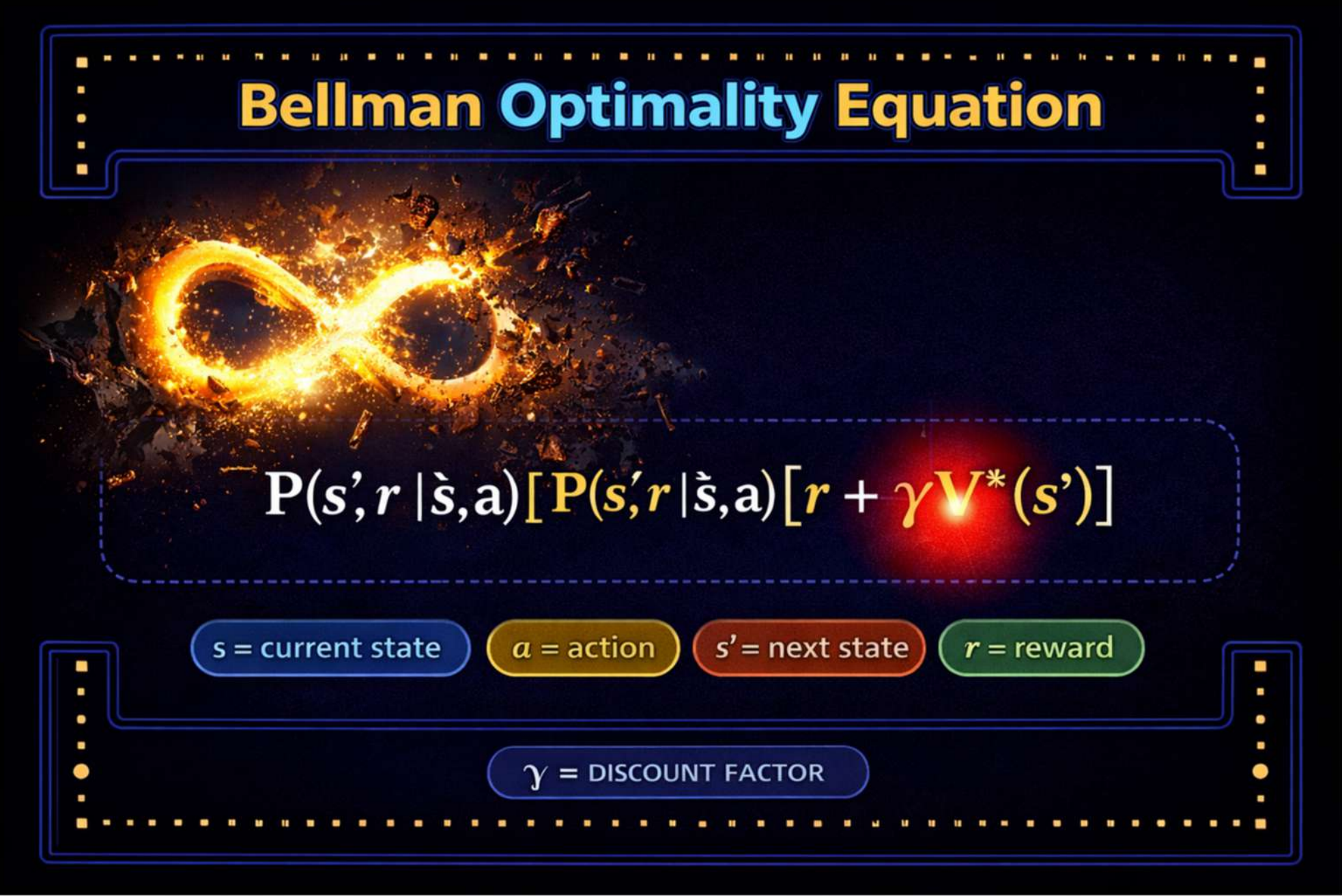
마시멜로와 벨만 방정식

3. 벨만 최적 방정식



마시멜로와 벨만 방정식

4. Gamma



The graphic features a dark background with a glowing orange infinity symbol in the center. The text 'Bellman Optimality Equation' is at the top in yellow and blue. Below the symbol is the equation $P(s', r | s, a) [P(s', r | s, a) [r + \gamma V^*(s')]]$. At the bottom, there are five colored boxes defining the variables: s = current state (blue), a = action (yellow), s' = next state (orange), r = reward (green), and γ = DISCOUNT FACTOR (blue).

Bellman Optimality Equation

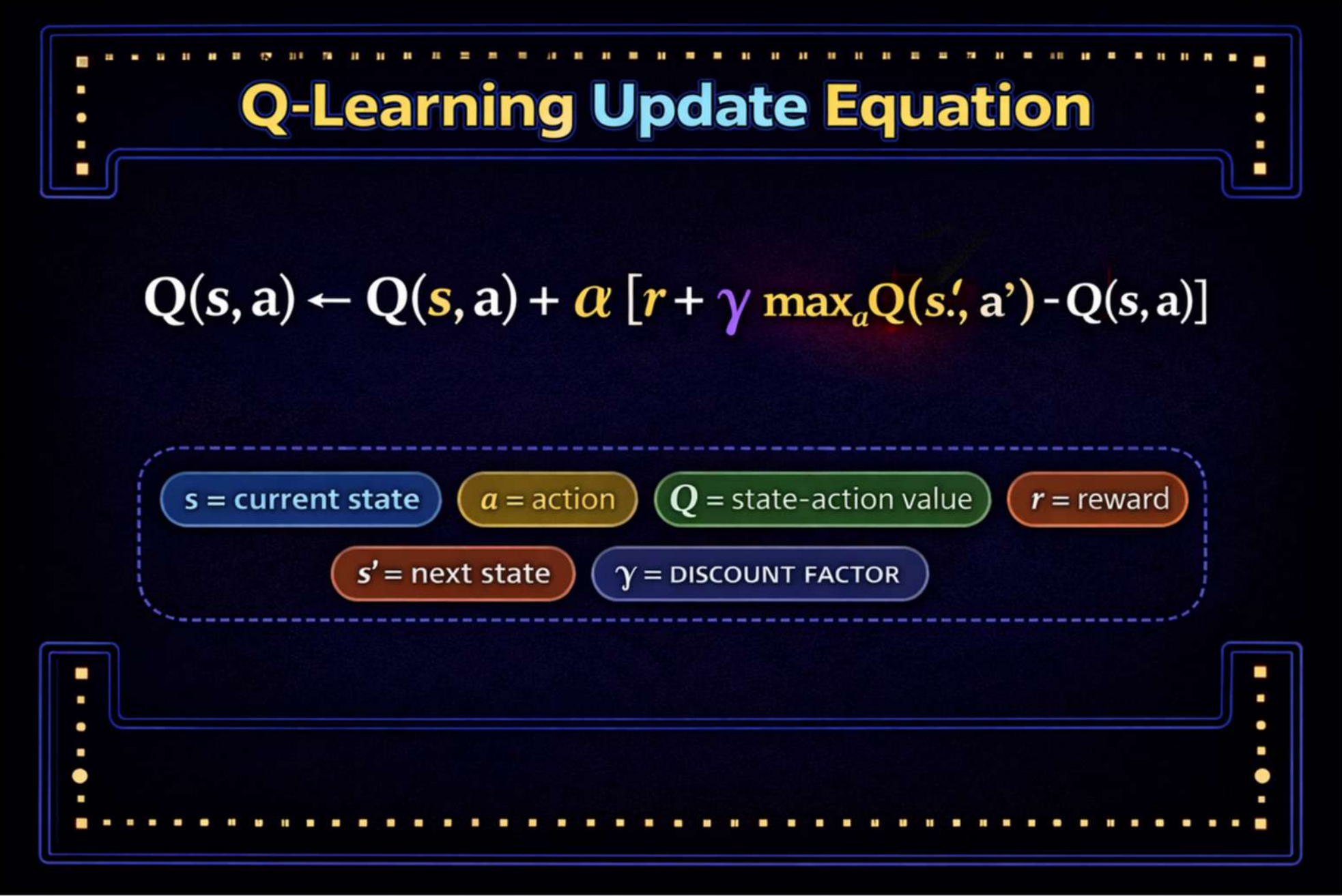
$$P(s', r | s, a) [P(s', r | s, a) [r + \gamma V^*(s')]]$$

s = current state a = action s' = next state r = reward

γ = DISCOUNT FACTOR

마시멜로와 벨만 방정식

5. 시간차 오차 (TD Error)



The diagram is titled "Q-Learning Update Equation" in a stylized font. Below the title is the equation: $Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_a Q(s', a') - Q(s, a)]$. The variables are color-coded: s is blue, a is yellow, α is orange, r is red, γ is purple, and s' is brown. Below the equation, there are five colored boxes with definitions: s = current state (blue), a = action (yellow), Q = state-action value (green), r = reward (red), s' = next state (brown), and γ = DISCOUNT FACTOR (purple). The entire diagram is enclosed in a blue border with yellow dots at the corners.

Q-Learning Update Equation

$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_a Q(s', a') - Q(s, a)]$$

s = current state a = action Q = state-action value r = reward
 s' = next state γ = DISCOUNT FACTOR

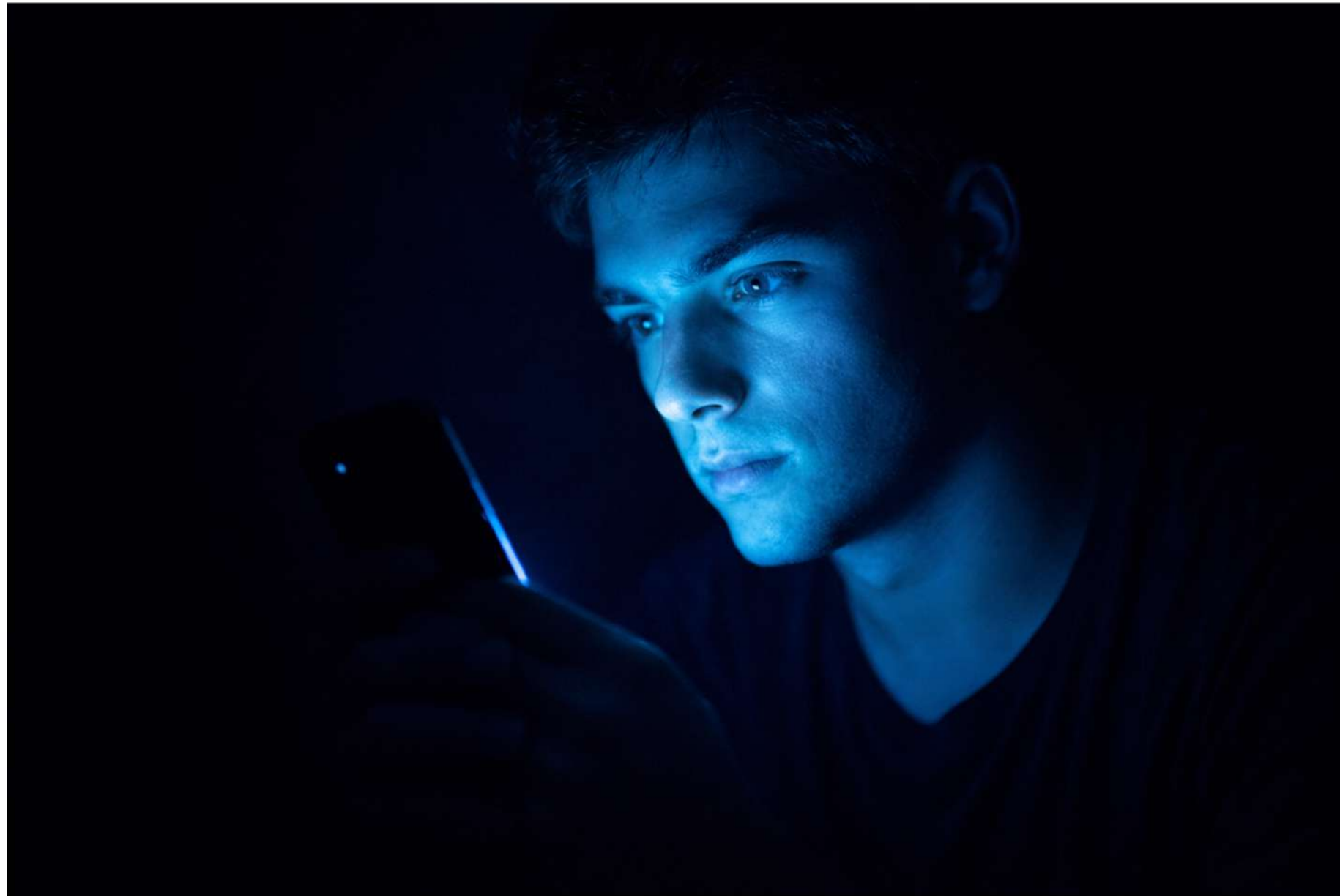
마시멜로와 벨만 방정식

6. 엡실론-그리디 (ϵ -greedy)



마시멜로와 벨만 방정식

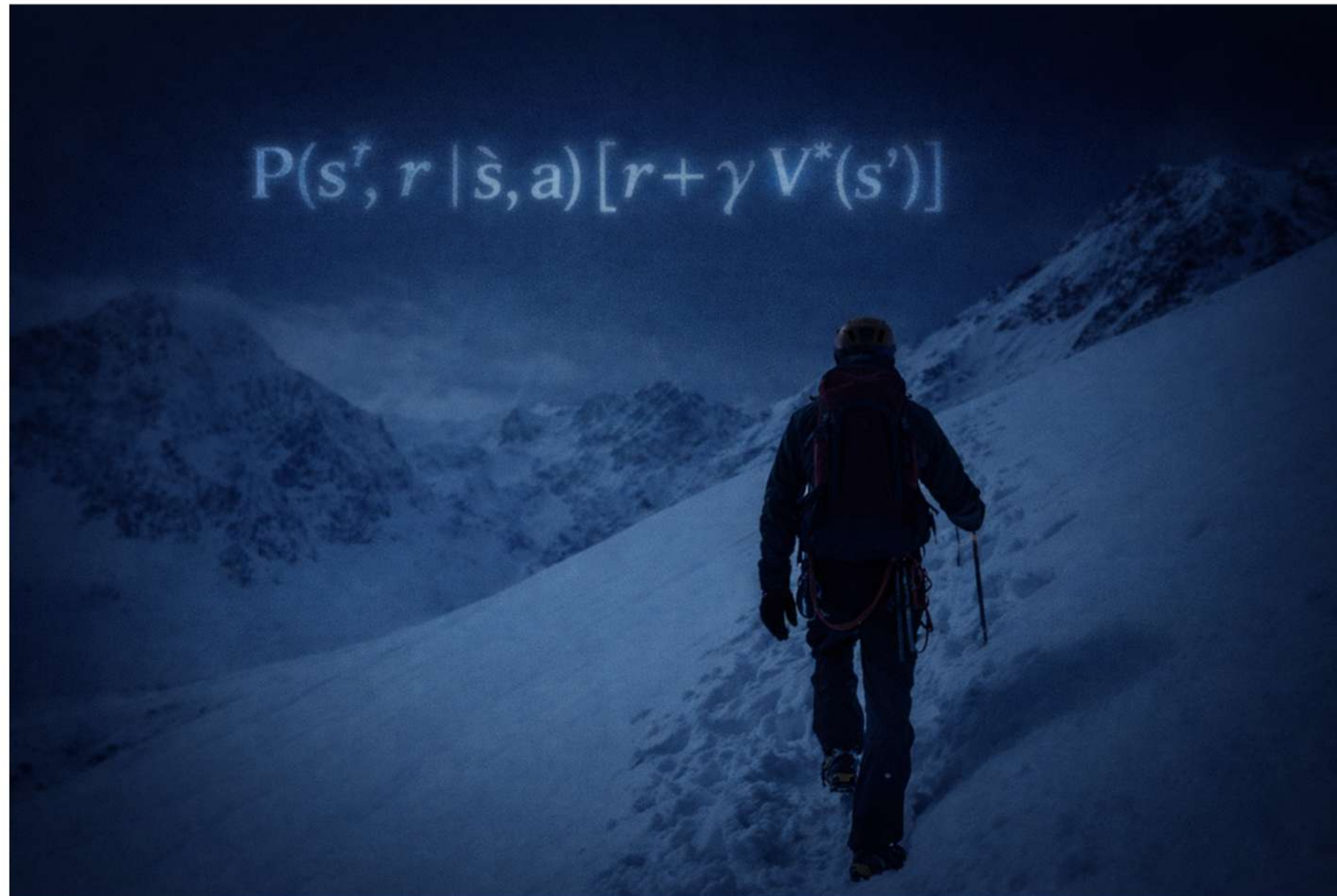
이주의..인사이트: 당신의 γ 는 얼마인가?



R L 은 어떻게 미래 기대가치를 현재로 가져오는가

마시멜로와 벨만 방정식

이주의..인사이트: 완벽한 에이전트가 되는 법



E.O.D